JAKHONGIR NODIROV

AI / Computer Vision Engineer

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SUMMARY

Computer Vision / AI engineer combining published research with full-cycle production experience — from model research through edge deployment. Design and deploy CNN and transformer based defect inspection systems running in production on AOI equipment for tier-1 electronics and semiconductor manufacturers. Comfortable owning a pipeline end to end: dataset design, model training, TensorRT/ONNX edge optimization, and production monitoring.

TECHNICAL SKILLS

- **Languages:** Python, C/C++
- **ML/DL:** PyTorch, PyTorch3D, TensorFlow, Scikit-Learn, NumPy, OpenCV
- **Specializations:** Object Detection/Segmentation/Tracking, Anomaly Detection, Medical Imaging, 3D Reconstruction, Generative/Diffusion Models
- **Edge Deployment:** TensorRT, ONNX, Jetson Nano, Docker/Compose, AWS EC2/S3, Oracle Cloud, CI/CD
- **Robotics/IoT:** ROS2, MicroROS, SLAM, Nav2, Raspberry Pi, ESP32, Yocto, PlatformIO

WORK EXPERIENCE

HyVision System Inc. Seongnam, South Korea
AI Research Software Engineer Jun 2025 - Present

- Own the full defect-inspection model lifecycle – dataset design, training, optimization, and on-device deployment – for AOI equipment used in semiconductor and consumer electronics manufacturing.
- Develop CNN based inspection models deployed in production, supporting high-precision, low false positive defect detection on the manufacturing line.
- Optimize trained models for edge inference (TensorRT/ONNX) to run efficiently on production hardware.

AIVAR Seoul, South Korea
AI Research Software Engineer Feb 2023 - Mar 2025

- Led development of a 3D human avatar reconstruction server application (3D morphable face/body models), deployed on AWS with Docker for consistent dev-to-production environments.
- Built and shipped a deep-learning image verification system (vector embedding search) for a government-backed digital IP-protection initiative.
- Researched and optimized CNN and transformer-based vision models across multiple concurrent product lines; owned deployment and production monitoring.

Gachon University — CT Lab Seongnam, South Korea
Graduate Researcher Sep 2020 – Feb 2023

- Designed a novel attention-based 3D U-Net architecture for volumetric MRI tumor segmentation, published in Sensors (2022) and IEEE ICTC (2021) – see Publications.
- Contributed to applied CV lab projects including road-condition assessment and license-plate recognition.

SELECTED PROJECTS

Semiconductor Wafer Defect Inspection *HyVision, 2025*

- Built a high-precision defect inspection pipeline for 8-inch and 12-inch silicon wafers using CNN-based object detection, segmentation, and anomaly detection models, deployed on industrial edge GPUs inside wafer inspection equipment.
- PyTorch, ONNX/TensorRT edge optimization · Deployed in production for continuous, high-precision defect detection with a low false-positive rate.

Consumer Electronics Surface Inspection *HyVision, 2025-2026*

- Built an end-to-end automated defect inspection pipeline (segmentation, detection, classification) for glass and metal enclosure parts, deployed on AOI machines in a tier-1 consumer electronics client's production line.
- PyTorch, ONNX/TensorRT edge optimization · Achieved fine-grained defect detection at production line speed with a low false-positive rate.

Driver Monitoring System (DMS) *Lab Project*

- Upgraded grayscale head-pose, facial landmark, and seatbelt-detection models using CNN and keypoint-based architectures; retrained on an augmented grayscale dataset and deployed via ONNX/TensorRT on NVIDIA Jetson for day-and-night reliability on edge hardware.

3D Human Avatar Reconstruction *AIVAR, 2023-2024*

- Converted 2D images into textured 3D face/body avatars using 3D morphable face and body model architectures, with diffusion-based image correction and neural texture-matching; deployed on AWS with Docker.

IP Protection — Image Verification *AIVAR, 2024*

- Built a vector-embedding image verification system for animation-character and logo IP protection using CNN-based face/feature recognition models, with a demo web application; part of a government-backed digital IP-protection project.

Attention 3D U-Net for Brain Tumor Segmentation *CT Lab, 2021-2022*

- Designed a novel attention-based 3D U-Net with multiple skip connections for volumetric MRI tumor segmentation, improving boundary-detection precision. Published in Sensors (2022).

Autonomous Driving Robot (ROS2) *Side Project*

- Built a mobile robot on Jetson Nano B1 with ROS2 Foxy, motor control, encoder feedback, RP-Lidar, and a web GUI (roslib.js) for remote operation; advanced SLAM/navigation capability.

EDUCATION

Gachon University 2020.09 – 2022.08
Master's Degree, Computer Engineering – GPA 4.17/4.5

PUBLICATIONS

- "Attention 3D U-Net with Multiple Skip Connections for Segmentation of Brain Tumor Images." *Sensors* 22, no. 17 (2022): 6501.
- "3D Volume Reconstruction from MRI Slices based on VTK." In *2021 International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 689–692. IEEE, 2021.

LANGUAGES

Uzbek (Native) · English (Proficient) · Korean (Intermediate)